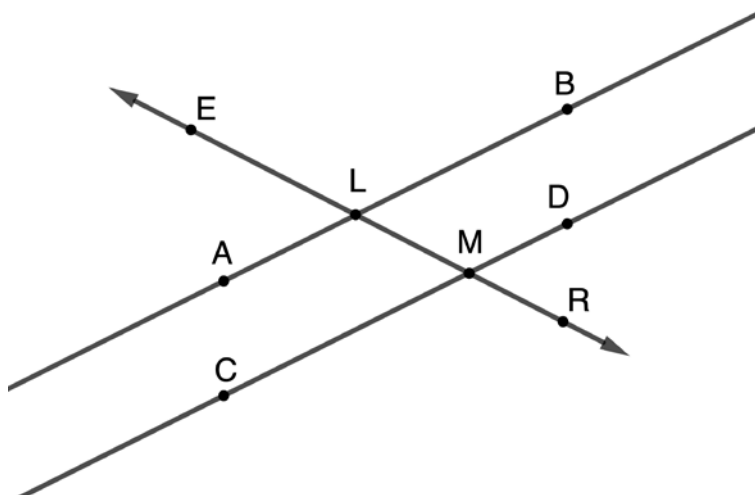


Geometry
Unit 3
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

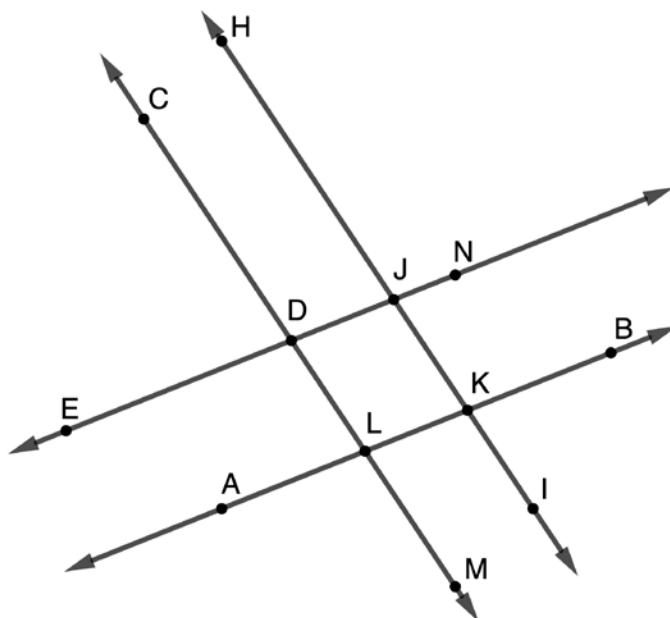
1. Three lines are shown, $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, and $\overleftrightarrow{ER}$. $\overleftrightarrow{ER}$ intersects $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$.



Determine if the given condition could be used to justify that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$?	Justification
If $m\angle ELA = 78^\circ$, then $m\angle DMR = 78^\circ$	☒ Yes ☐ No	If two lines are intersected by a transversal so the alternate exterior angles are congruent, then the lines are parallel.
$\angle ALM \cong \angle CMR$	☒ Yes ☐ No	If two lines are intersected by a transversal so the corresponding angles are congruent, then the lines are parallel.
$m\angle ELA + m\angle ELB = 180^\circ$	☐ Yes ☒ No	Two supplementary adjacent angles do not form a set of two parallel lines.
$\angle ELB \cong \angle CMR$	☒ Yes ☐ No	If two lines are intersected by a transversal so the alternate exterior angles are congruent, then the lines are parallel.
$\angle LMC \cong \angle DMR$	☐ Yes ☒ No	Congruent vertical angles do not prove parallel lines
If $m\angle ELB = 95^\circ$, then $m\angle ELA = 85^\circ$	☐ Yes ☒ No	Two supplementary adjacent angles do not form a set of two parallel lines.
If $m\angle DMR = 81^\circ$, then $m\angle ALM = 99^\circ$	☒ Yes ☐ No	Since $m\angle DMR + m\angle CMR = 180^\circ$ because they form a linear pair, then $m\angle CMR = 99^\circ$. Since $m\angle ALM = 99^\circ$ and $m\angle CMR = 99^\circ$, $m\angle ALM = m\angle CMR$ by transitive property, and $\angle ALM \cong \angle CMR$ by the definition of congruence and since the corresponding angles are congruent, the lines intersected by the transversal must be parallel.

2. $\overleftrightarrow{CM}$, $\overleftrightarrow{HI}$, $\overleftrightarrow{EN}$, and $\overleftrightarrow{AB}$ are shown, where $m\angle CDE = 90^\circ$.



Determine if the given conditions could be used to justify that $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$.

Condition	Is $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$?	Justification
$\angle CDE \cong \angle HJN$	☐ Yes ☒ No	If two lines are intersected by a transversal so the consecutive exterior angles are congruent, that is not enough information to justify congruence.
$\angle CDJ \cong \angle DLK$	☒ Yes ☐ No	If two lines are intersected by a transversal so the corresponding angles are congruent, then the lines are parallel.
$m\angle EDL + m\angle NJK = 180^\circ$	☒ Yes ☐ No	If two lines are intersected by a transversal so the consecutive exterior angles are supplementary, then the lines are parallel.
$\angle KJD \cong \angle HJN$	☐ Yes ☒ No	Congruent vertical angles do not prove that two lines are parallel.
$\angle JKL \cong \angle BKI$	☐ Yes ☒ No	Congruent vertical angles do not prove that two lines are parallel.
$m\angle BKI + m\angle LKI = 180^\circ$	☐ Yes ☒ No	Line pair of angles do not prove that two lines are parallel.